

Black-Box Sparse Adversarial Attack on EEG Recognition

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Dataset: BNCI2014-001

(BCI Competition IV, Dataset 2a)

Subjects	9 healthy participants
EEG channels	22 (Ag/AgCl, 3.5 cm spacing)
Sampling rate	250 Hz $\rightarrow$ resampled 128 Hz
Bandpass	4–38 Hz
Sessions	2 per subject (different days)
Trials	288 per session per subject
Trial length	4 s $\Rightarrow X \in \mathbb{R}^{22 \times 512}$

Motor imagery paradigm:

Each trial = one imagined movement

left hand, right hand, feet, tongue

(4 classes, 12 trials each per run, 6 runs per session)

Our task: subject identification

Given an EEG trial, predict **which of the 9 subjects** recorded it.

Classes	9 (one per subject)
Train	session 0 (all 9 subjects)
Test	session 1 (different day)
Protocol	cross-session

Why this task?

Subject ID from EEG is used in biometric authentication systems $\Rightarrow$ a natural target for an impersonation attack.

Brunner et al., 2008; MOABB library

Biometric Evaluation: FAR, FRR, and EER

Verification view: for a claimed identity c , accept iff the score $s_c(X) = f_c(X)$ exceeds threshold τ .

False Accept Rate (FAR):

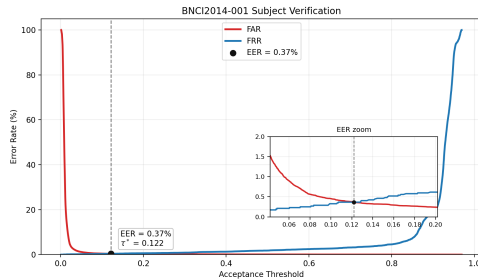
$$\text{FAR}(\tau) = \frac{\#\{s_c(X_{\text{imp}}) \geq \tau\}}{N_{\text{imp}}}$$

False Reject Rate (FRR):

$$\text{FRR}(\tau) = \frac{\#\{s_y(X_{\text{gen}}) < \tau\}}{N_{\text{gen}}}$$

Equal Error Rate (EER):

$$\tau^* : \text{FAR}(\tau^*) = \text{FRR}(\tau^*), \quad \text{EER} = \text{FAR}(\tau^*) = \text{FRR}(\tau^*)$$



Clean cross-session subject verification: pooled
EER = 0.37%

EEGConformer: Architecture and Threat

Input: $X \in \mathbb{R}^{22 \times 512}$ (22 ch $\times$ 4 s at 128 Hz)

Architecture:

1. Temporal conv — local oscillation patterns
2. Spatial conv — cross-channel mixing
3. Average pooling — dimensionality reduction
4. Self-attention ($N=6$ layers, 10 heads)
global temporal dependencies
5. FC classifier $\rightarrow$ 9 subject scores $f_k(X)$

Training: cross-entropy, Adam, 100 epochs

Alignment: Euclidean alignment per subject

Song et al., IEEE TNSRE 2023

[EEGConformer diagram]

$X \in \mathbb{R}^{22 \times 512} \rightarrow \text{Conv} \rightarrow \text{Attention} \rightarrow$
 $f_k(X)$
9 subject scores

Output: $f_k(X)$ for $k = 1, \dots, 9$

One score per subject.

The predicted identity is $\hat{y} = \arg \max_k f_k(X)$.

EEG Classification Model and Security Threat

The signal: EEG trial $X \in \mathbb{R}^{C \times T}$

C electrodes, T samples at rate f_s

The model: EEGConformer

Temporal conv $\rightarrow$ spatial conv $\rightarrow$ self-attention

Output: class scores $f_k(X)$ (logits)

The task: EEG-based user identification

9 subjects, train/test split

The threat: attacker sees **only** the output scores

No model weights. No gradients. No training data.

[EEGConformer diagram]

Input $X \rightarrow$ Conv $\rightarrow$ Attention $\rightarrow$
logit

Song et al., IEEE TNSRE 2023

EEG Classification Model and Security Threat

[EEGConformer diagram with orange feedback loop]

$X + E \rightarrow \text{Model} \rightarrow \text{logit} \rightarrow \text{Attacker reads score}$
 $\leftarrow \text{😊 crafts new } E$

**Can an unauthorized user
bypass EEG authentication
through a black-box attack?**

What is an Adversarial Attack?

Classic example — images:

$$\begin{array}{ccccc} \boxed{\text{Panda}} & + & \text{invisible noise} & = & \boxed{\text{Gibbon?}} \\ 57\% \text{ panda} & & & & 99\% \text{ gibbon} \end{array}$$

The model changed its mind completely based on a signal **invisible to humans**.

Our setting — EEG:

$$\begin{array}{ccccc} \boxed{\text{Brain signal}} & + & \text{fake wave} & = & \boxed{\text{Wrong ID?}} \\ \text{Subject A} & & & & \text{Subject B?} \end{array}$$

Same question: add a tiny perturbation, make the classifier disagree.

Key difference: the perturbation must look like a real brain wave.

Blackbox Sparse Adversarial Attack

⇒ **Adversarial attack:** add perturbation E to X so the model outputs the wrong label
 $\hat{y}(X + E) \neq y(X)$

⇒ **Sparse:** only a few electrode channels are perturbed

$$\gamma(E) = |\{c : E_{c,:} \neq 0\}| \leq B \quad (\text{e.g. } B = 1, 2, 3 \text{ out of } 22)$$

Why? In a real system, an attacker can only interfere with a limited number of electrode contacts — not rewire the whole headset.
Fewer channels also means harder to detect.

⇒ **Black-box:** no model weights, no gradients — score queries only

Why? In a deployed authentication system the attacker submits a signal and gets back “accepted / rejected” or a confidence score. Nothing more.

⇒ **Stealthy:** perturbation amplitude is bounded

$$\|E\|_{\infty} \leq \epsilon \quad (\text{less than } 10\% \text{ of signal peak amplitude})$$

~~Blackbox Sparse Adversarial Attack — Generic Problem~~

⇒ $f_k(\mathbf{X})$: score for class k , with $k = 1, \dots, K$

⇒ Adversarial example: $\mathbf{X}^{\text{adv}} = \mathbf{X} + \mathbf{E}$

Generic objective: change the model decision while perturbing as few channels as possible and keeping the amplitude bounded.

$$\begin{aligned} \min_{\mathbf{E} \in \mathbb{R}^{C \times T}} \quad & \gamma(\mathbf{E}) \\ \text{s.t.} \quad & \arg \max_{k=1, \dots, K} f_k(\mathbf{X} + \mathbf{E}) \neq \arg \max_{r=1, \dots, K} f_r(\mathbf{X}) \\ & \|\mathbf{E}\|_{\infty} \leq \epsilon \end{aligned}$$

where $\gamma(\mathbf{E}) = \sum_{c=1}^C 1[\mathbf{E}_{c,:} \neq 0]$ counts attacked channels.

This is the **generic, model-agnostic** attack problem. Next we rewrite the success condition in a form that is easier to optimize.

Clean predicted class:

$$\hat{y} = \arg \max_{r=1,\dots,K} f_r(X)$$

Score used during attack design:

$$\Gamma(X + E) = \max_{j \neq \hat{y}} f_j(X + E) - f_{\hat{y}}(X + E)$$

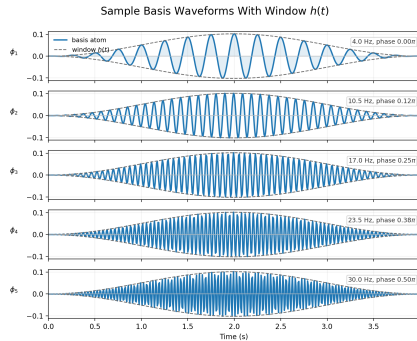
- ⇒ **Interpretation:** compare the best competing class against the clean winner $\hat{y}$, both evaluated on the perturbed signal.
- ⇒ **Success criterion:** the attack succeeds exactly when $\Gamma(X + E) \geq 0$.
- ⇒ This scalar score is what we try to increase while designing E: larger Γ means closer to flipping the decision.

Designing E for EEG

⇒ **Design principle:** on each attacked channel, E should look like a small EEG-like oscillatory waveform.

⇒ **Channel-wise model:** if channel c is attacked, we inject a waveform δ_c built from physiological atoms.

$$\delta_c(t) = \sum_{i=1}^M a_{c,i} \phi_i(t), \quad \phi_i(t) = h(t) \cos\left(\frac{2\pi f_i t}{f_s} + \psi_i\right)$$



Example basis atoms $\phi_i(t)$ with window $h(t)$

$$[\mathbf{e}_c]_{c,:} = \delta_c^\top, \quad [\mathbf{e}_c]_{i,:} = \mathbf{0}^\top \ (i \neq c), \quad \mathbf{E} = \sum_{c=1}^C \mathbf{e}_c$$

$$\gamma(\mathbf{E}) = |\{c : \delta_c \neq \mathbf{0}\}|$$

~~EEG-Constrained Problem We Actually Solve~~

Combining the attack score with the channel-wise waveform model gives

$$\begin{aligned} \min_{\mathcal{S}, \{a_{c,i}\}} & |\mathcal{S}| \\ \text{s.t. } & \Gamma\left(\mathbf{X} + \sum_{c \in \mathcal{S}} \mathbf{e}_c\right) \geq 0 \\ & \left\| \sum_{c \in \mathcal{S}} \mathbf{e}_c \right\|_{\infty} \leq \epsilon \end{aligned}$$

where $\mathcal{S}$ is the attacked channel set, $\delta_c(t) = \sum_{i=1}^M a_{c,i} \phi_i(t)$, and $\mathbf{e}_c$ is the perturbation matrix with only row c nonzero.

This is the **actual EEG attack problem**: sparse channel selection, waveform coefficients, attack success, and amplitude bound all in one place.

What the Perturbation Looks Like

Clean channel: real EEG trace
irregular oscillations, noisy

Injected waveform: $\delta_c(t)$
smooth low-frequency mixture, small amplitude

Perturbed channel: $x_c + \delta_c$
visually almost identical to clean

To a human reader:
the two traces look nearly identical

To the classifier:
the perturbed signal may cross the
decision boundary

This is the key intuition:
we are not adding arbitrary noise;
we are injecting a small EEG-like
waveform
on a small set of channels.

[plot: clean trace / injected wave / perturbed
trace]

How We Solve the EEG-Constrained Problem

Why this is hard: choosing the channels to attack is combinatorial. With 22 channels, a budget of B gives $\binom{22}{B}$ candidates.

Solution strategy — two phases:

Phase 1

Channel scoring

Which channels enter $\mathcal{S}$?



Phase 2

Waveform amplitude tuning

Which coefficients $a_{c,i}$ on that support?

Phase 1 — Channel Scoring

Idea: first score channels one by one, before doing any expensive optimization.

- ⇒ Pick one channel c .
- ⇒ Generate a few small EEG-like **probe waveforms** on that channel only, using random frequencies in the allowed band.
- ⇒ For each probe, query the model and measure how much the classifier scores change.
- ⇒ Average those score changes to get one **channel score** $\bar{\Gamma}_c$.
- ⇒ Repeat this for all 22 channels.

Channel Selection

Rank the channels by their average score and keep the top- B channels as the candidate subset $\mathcal{S}$ for Phase 2.

Phase 2 — ~~Waveform Amplitude Tuning~~

After Phase 1, keep only the selected channels and choose how strong the attack waveform should be.

- ⇒ **Goal:** find the smallest attacked time span on those channels that can still launch the attack.
- ⇒ Start with a short attacked segment and a small waveform amplitude.
- ⇒ Tune a scaling factor for the waveform and query the model.
- ⇒ If the attack is still too weak, increase the attacked segment size and tune the amplitude again.
- ⇒ Stop when the estimated attack success is high enough, while staying inside the amplitude budget.
- ⇒ In practice, we implement this iterative tuning with **SPSA**.

What Makes an Attack *Stealthy*?

Goal: the adversarial signal should be hard to detect and hard to filter out.

Three concrete questions a defender asks:

⇒ **Does the perturbation break the spatial structure of EEG?**

⇒ measured by the **covariance distortion** $\|\text{Cov}(X^{\text{adv}}) - \text{Cov}(X)\|_F$

Real EEG channels are correlated; a big shift is easy to flag.

⇒ **How many electrodes are actually touched?**

⇒ measured by **channel sparsity** $\gamma(E)/C$, the fraction of active channels

Attacking fewer electrodes limits both visibility and damage.

⇒ **Does the frequency content of the signal change?**

⇒ measured by **PSD deviation** $|\overline{\text{PSD}}_{\text{adv}} - \overline{\text{PSD}}_{\text{clean}}|$ (dB)

Unnatural spectral peaks are a clear fingerprint of injection.

A stealthy attack achieves misclassification *without* triggering any of these detectors.

Stealthiness Comparison: Ours vs. Filter_Attack

What We Compare Against

Filter_Attack (Meng et al., *Information Fusion* 2024) is a **white-box spatial-filter baseline**: it learns a dense channel-mixing matrix $\mathbf{x}^{\text{adv}} = (\mathbf{W} + \mathbf{I})\mathbf{x}$ and perturbs *all* channels jointly. Our method instead is **black-box** and **channel-sparse**.

Metric	Ours	Filter_Attack
Attack Success Rate	100%	100%
Channels perturbed	18.2%	100%
$\ \mathbf{E}\ _2$ (signal amplitude)	5.31	28.58
$\ \mathbf{E}\ _\infty$	0.64	1.12

- ⇒ Same attack success rate — but our covariance distortion is 70× smaller.
- ⇒ PSD deviation 62× smaller: the injected waveform is confined to physiological frequencies by construction.
- ⇒ Filter_Attack is white-box: ours uses score-only access

Summary

- ⇒ **Threat model:** score-only access, sparse channel injection, bounded amplitude
⇒ realistic, stealthy, and physically plausible
- ⇒ **Formulation:** choose the smallest attacked channel set whose channel-wise waveform perturbations make the attack score non-negative under the amplitude bound
⇒ sparse support selection + EEG-constrained waveform coefficients
- ⇒ **Waveform basis:** frequency constraint by construction, not penalty
⇒ every perturbation looks like a real neural oscillation
- ⇒ **Algorithm:** channel scoring + waveform amplitude tuning
⇒ black-box, query-efficient, principled

One-sentence version:

Score channels to find a small vulnerable subset, inject a small mixture of physiologically-realistic oscillations, tune the mix until the classifier breaks.